

PAUL MITCHELL

CYBERSECURITY & SYSTEMS ANALYSIS | IT INFRASTRUCTURE & AI/ML

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PROFESSIONAL SUMMARY

William & Mary Computer Science M.S. candidate (3.95 GPA, AI/ML concentration) and B.S. graduate providing departmental technical support and Linux systems administration across 100+ instructional machines. Combines hands-on IT infrastructure management and cloud telemetry with cybersecurity project engineering, including adversarial prompt defense proxies and network anomaly detection. Proven ability to translate complex systems data into operational runbooks, actionable executive metrics, and audit-ready governance frameworks.

EDUCATION

College of William & Mary

Williamsburg, VA

Master of Science in Computer Science (AI/ML Concentration) | Expected Dec 2026 | GPA: 3.95

Bachelor of Science in Computer Science | Graduated Dec 2024

Relevant Coursework: Advanced Software Engineering, Machine Learning Systems, Deep Learning, Data Mining

TECHNICAL SKILLS

Systems & Infrastructure: Linux (Ubuntu/RHEL), Windows, SSH, DNS, static IPs, Netplan, NFS, Docker, Puppet, system hardening

Monitoring & Operations: Prometheus, Grafana, dashboard engineering, alert rule design, log auditing, capacity planning

Programming & Scripting: Python, Bash, PowerShell, Java, SQL, YAML, Git/GitLab CI, FastAPI, RESTful APIs

Security & Systems Analysis: Threat assessment, technical runbooks, anomaly detection, threshold calibration, root-cause analysis, Excel

TECHNICAL & PROFESSIONAL EXPERIENCE

IT Administrator | William & Mary Technology Support Center | Williamsburg, VA

Feb 2025 - Present

- Administer, troubleshoot, and support more than 100 Linux instructional systems, resolving network connectivity, routing, DNS, account, and software issues for campus users.
- Built Prometheus and Grafana telemetry dashboards tracking GPU compute clusters, storage health, and node uptime, accelerating incident response and issue resolution.
- Conducted architectural evaluation of multi-user Docker containerization, documenting security isolation, resource overhead, and operational maintenance trade-offs for leadership.
- Standardized network configurations and maintained structured inventory records in Excel, eliminating configuration drift across multi-room compute environments.
- Developed automated Bash recovery scripts for stalled network daemons, eliminating recurring manual administrator interventions.

Data Analytics & Engineering Extern | Extern, Amazon-Sponsored Project | Remote

Jun 2025 - Sep 2025

- Engineered reproducible Python and AWS S3 data pipelines to clean and structure ~20,000 records, cutting recurring data prep cycles from hours to minutes.
- Built an interactive Streamlit analytics application that synthesized complex model outcomes into actionable executive decision-support metrics for program mentors.
- Evaluated predictive modeling trade-offs using LightGBM and NLP sentiment clustering, presenting strategic recommendations directly to stakeholders.
- Automated dataset versioning and storage reconciliation using DVC on AWS S3 to maintain audit-ready experimental lineages.

KEY TECHNICAL & SYSTEMS PROJECTS

Real-Time AI Security & Prompt Injection Defense Proxy | Python, FastAPI

2026

- Architected pre-inference proxy middleware analyzing token structures to intercept adversarial prompts before model execution; benchmarked scoring latency at 0.038ms to 1.11ms.
- Documented enterprise integration patterns and security trade-offs for production LLM deployments, eliminating need for secondary verification model calls.

Enterprise IT Asset & Security Configuration Tracker | Python, Excel

2026

- Built an asset governance system tracking configuration drift, documentation gaps, and patch compliance across 100+ production instructional hosts.
- Designed automated normalization workflows for hardware lifecycles and software licenses, establishing audit-ready record maintenance procedures.

Healthcare Network Anomaly Detection Engine | Python, scikit-learn

2025

- Designed network telemetry inspection workflows featuring automated feature extraction and threshold calibration to support SOC operational alerting and threat triage.
- Translated complex traffic variance models into operational runbooks and decision matrices for technical support personnel.